Plant Hormones

Subject & topics: Biology | Physiology | Homeostasis & Response **Stage & difficulty:** A Level P2

Part A

Hormone functions

Match the hormones to the functions in the table below.

Hormone	Function(s)
	promotes cell elongation, represses branching, regulates tropisms
	promotes cell division
	promotes seed germination and stem elongation
	promotes fruit ripening and leaf abscission
	prevents growth of seeds and buds during winter, and closes stomata in low water conditions

Items:

cytokinins | gibberellins | abscisic acid (ABA) | ethene/ethylene | auxins

Part B

Dwarf plants

A crop breeder wants to breed a shorter crop. What are some advantages of having shorter crops? Select all that apply.

- [ ] Plants divert more nutrient resources to the fruits/ grains.
- [ ] Plants are less prone to falling over.
- [ ] Plants can harvest more light energy.
- [ ] Plants are more drought-tolerant.
- [ ] Plants can be grown closer together.

Which of the following could be effective strategies for producing shorter plants? Select all that apply.

- [ ] boosting gibberellin signalling
- [ ] repressing gibberellin signalling
- [ ] boosting ABA signalling
- [ ] repressing ABA signalling
- [ ] boosting ethene signalling
- [ ] repressing ethene signalling

Which of the following could be an unwanted effect of pursuing this strategy? Select all that apply.

- [ ] poor drought tolerance
- [ ] increased branching
- [ ] overheating
- [ ] delayed fruit ripening
- [ ] poor seed germination
- [ ] reduced carbon fixation

Part C

Drought tolerance

A different crop breeder wants to breed a more drought-tolerant crop.

Which of the following could be effective strategies for producing drought-tolerant plants? Select all that apply.

- [ ] boosting auxin signalling
- [ ] repressing auxin signalling
- [ ] boosting ABA signalling
- [ ] repressing ABA signalling
- [ ] boosting ethene signalling
- [ ] repressing ethene signalling

To pursue this strategy, the crop breeder planted 7 different crop varieties (A-G). They grew each variety alongside their current variety (labelled **control**) and imaged them with a thermal imaging camera. Their results are shown in **Figure 1**.

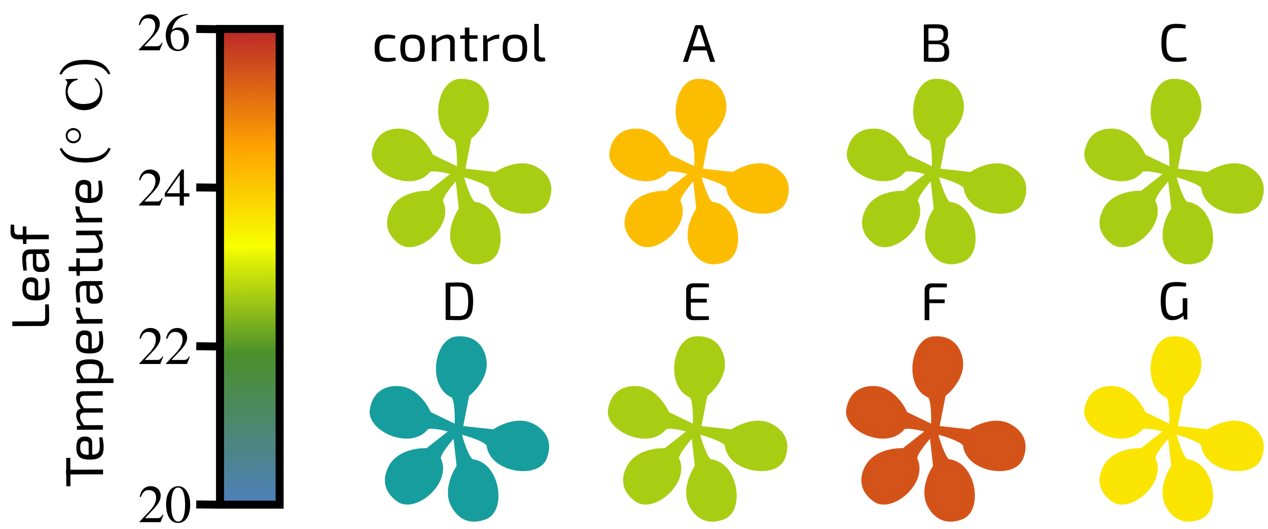


Figure 1: Leaf temperature in different plant varieties.

Which varieties should the breeder use in their breeding programme? Select all that apply.

- [ ] A
- [ ] B
- [ ] C
- [ ] D
- [ ] E
- [ ] F
- [ ] G

Which of the following could be an unwanted effect of pursuing this strategy? Select all that apply.

- [ ] increased branching
- [ ] poor seed germination
- [ ] increased plant height
- [ ] delayed fruit ripening
- [ ] overheating
- [ ] reduced carbon fixation

Question deck:

STEM SMART Biology Week 21 - Plant Growth & Reproduction

isaac You work it out.

STEM SMART Biology Week 21 - Plant Growth & Reproduction >

Phototropism and Gravitropism

Subject & topics: Biology | Physiology | Homeostasis & Response **Stage & difficulty:** A Level P2

In most plants, the shoots exhibit positive phototropism (i.e. they grow towards the light) and the roots exhibit positive gravitropism (i.e. they grow towards the centre of the Earth). Both of these growth responses are regulated by indoleacetic acid (IAA), which belongs to a group of plant hormones called auxins.

Part A

Phototropism

Indoleacetic acid (IAA) is mostly produced in the ____, where high levels of IAA ____ growth. This ensures that the plant grows upwards.

High light intensity causes IAA to be transported ____ the light, causing the cells on the ____ to elongate more. This causes the stem to bend ____ the direction of light.

Items:

xylem vessels | apical bud | stimulate | inhibit | towards | away from | same side as the light | opposite side from the light

Part B

Gravitropism

In the roots, high levels of IAA ____ growth and low levels of IAA ____ growth.

IAA flows down into the roots, which ensures the roots grow downwards. If a root does not run parallel to the direction of gravity, higher levels of IAA will accumulate on the bottom side. This will ____ elongation of cells on this side. The low levels of IAA on the upper side will ____ elongation of cells on that side. This will cause the root to bend ____ the centre of the Earth.

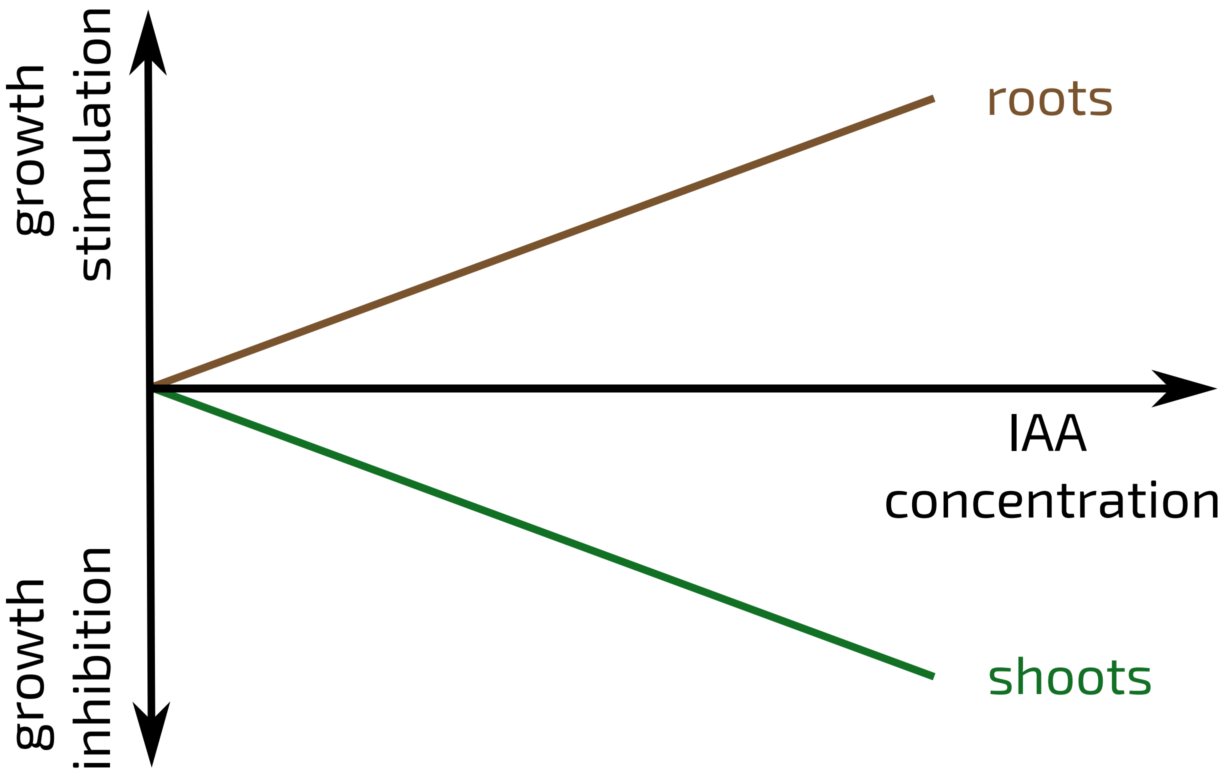
Items:

stimulate | inhibit | towards | away from

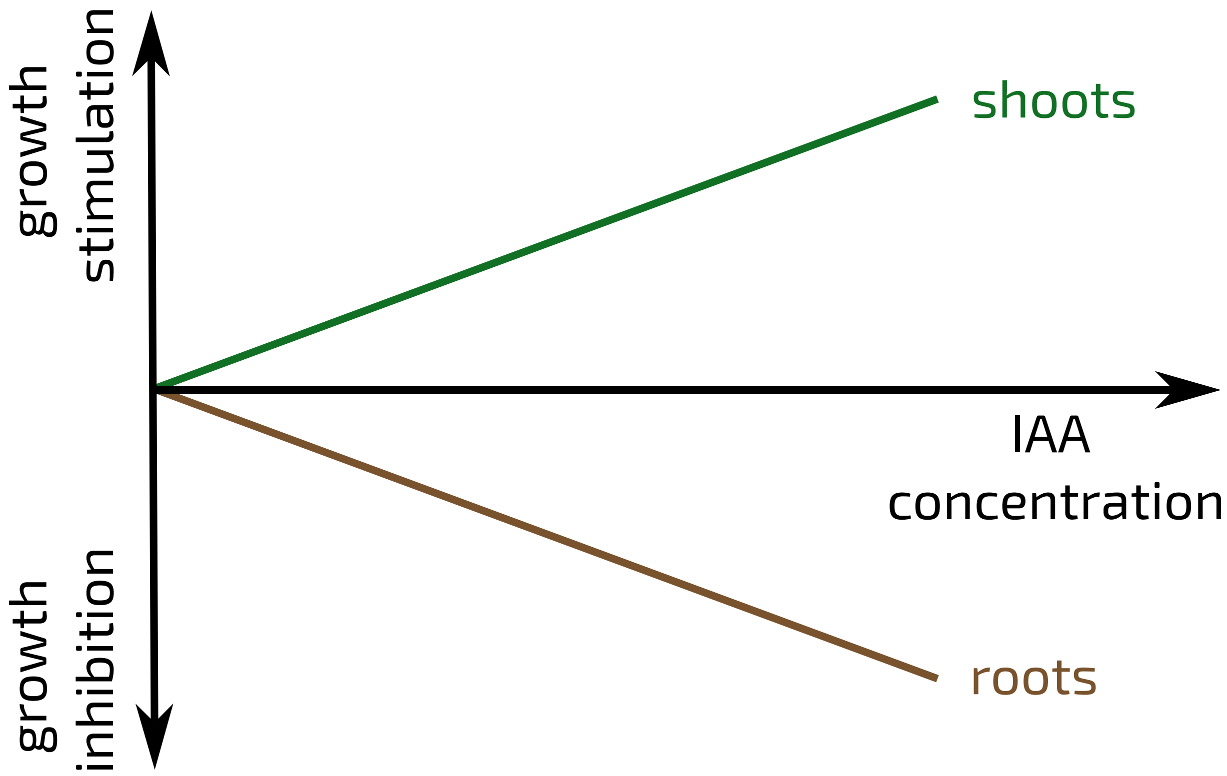
Part C

Roots vs shoots

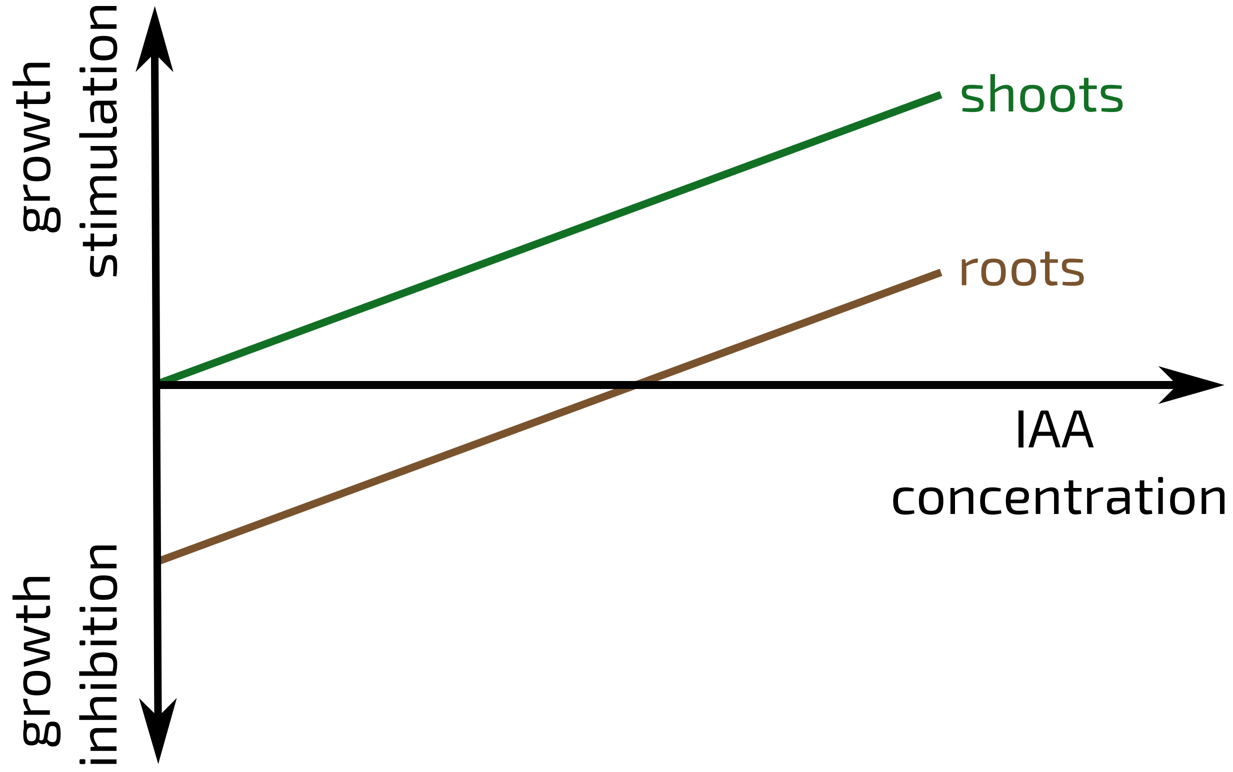
Which of the graphs below correctly shows the effects of different concentrations of indoleacetic acid (IAA) on roots and shoots?



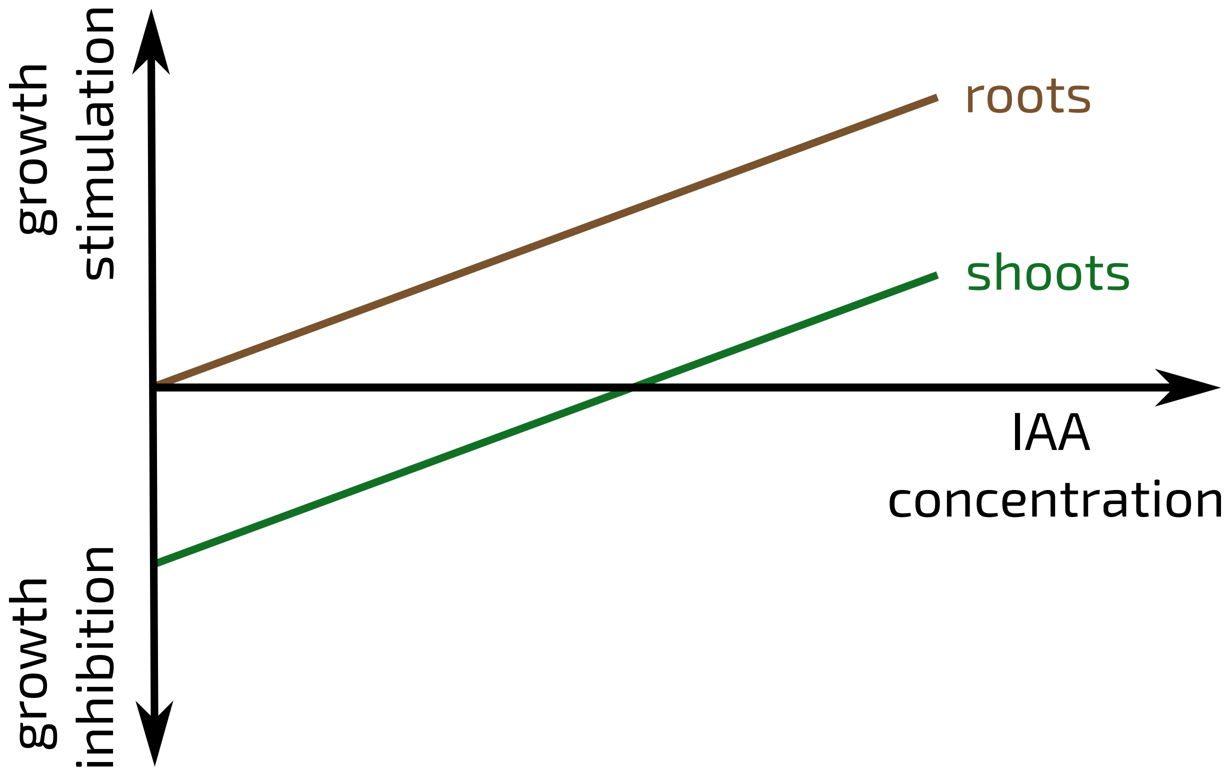
A



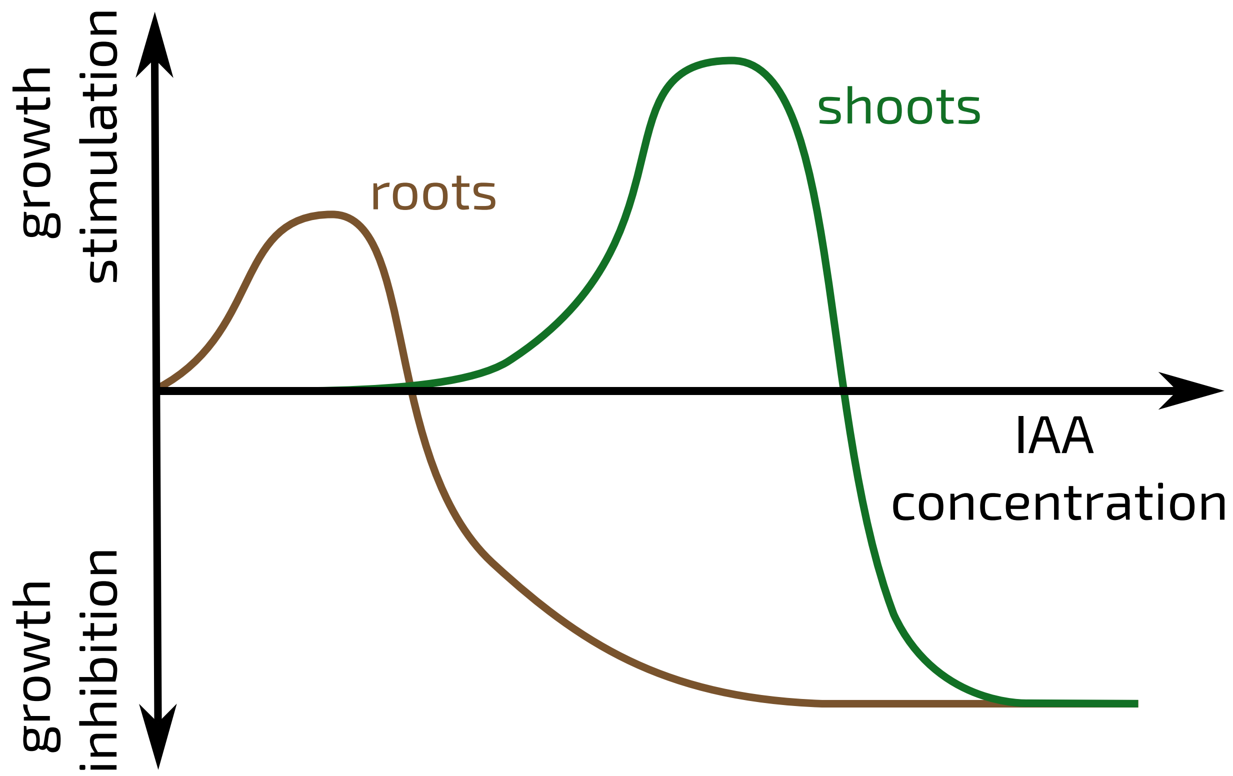
B


growth stimulation
shoots
roots
IAA concentration
growth inhibition

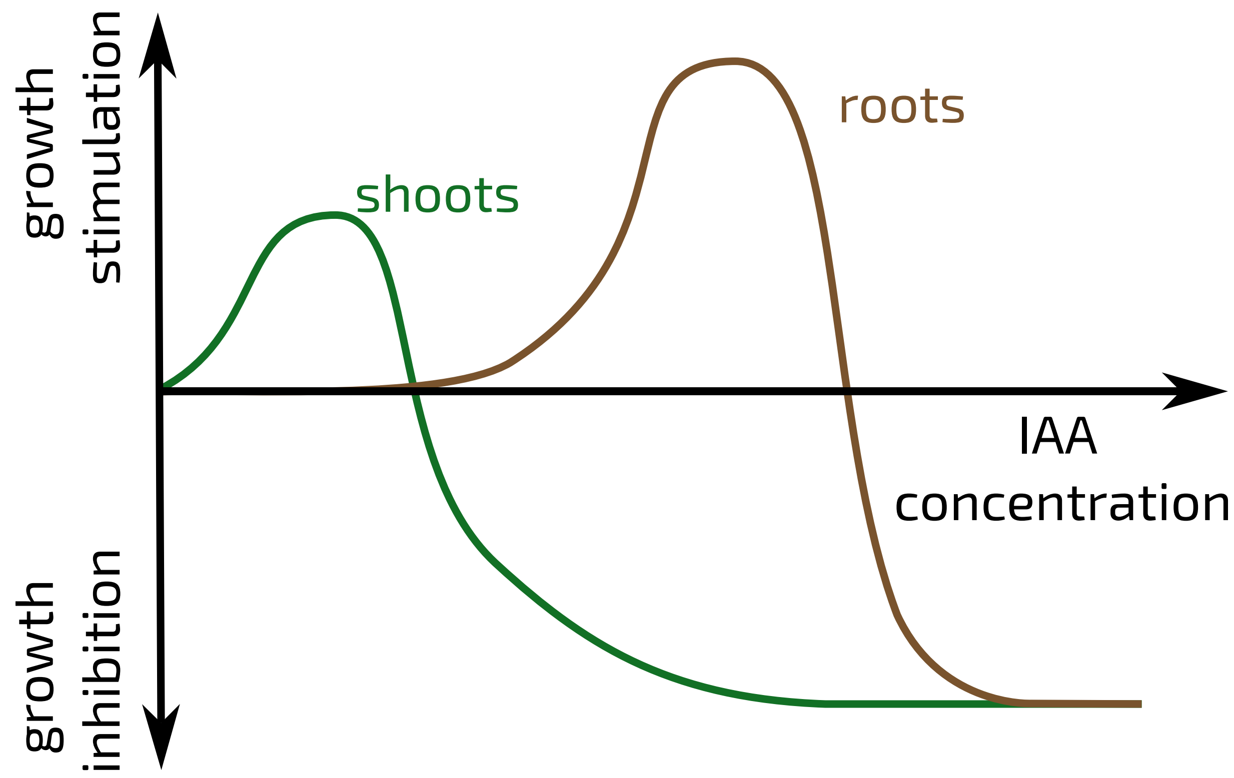
C


growth stimulation
roots
shoots
IAA concentration
growth inhibition

D


growth stimulation
shoots
roots
IAA concentration
growth inhibition

E



F

- Graph A
- Graph B
- Graph C
- Graph D
- Graph E
- Graph F

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Question deck:

STEM SMART Biology Week 21 - Plant Growth & Reproduction

Plants in Space

Subject & topics: Biology | Physiology | Homeostasis & Response **Stage & difficulty:** A Level P2

All plants on earth grow in the presence of gravity. There are various techniques that biologists can use to try and alter/reduce the effects of gravity on plants (e.g. rotating a plant continuously, or mutating genes that are important for sensing gravity), but none of these remove the effect of gravity completely. A group of biologists enlisted the help of some astronauts on the International Space Station (ISS), to study how plants grow in microgravity [1].

The astronauts and their ground-based colleagues germinated seedlings and tracked their growth over several days. The seedlings were germinated in square petri dishes containing solid (agar-based) growth medium. The petri dishes were placed "vertically" inside growth chambers. Bright LED lights were situated at the "top" of the chambers (**Figure 1**). The growth chambers precisely controlled the environmental conditions experienced by the seedlings on earth and on the ISS.

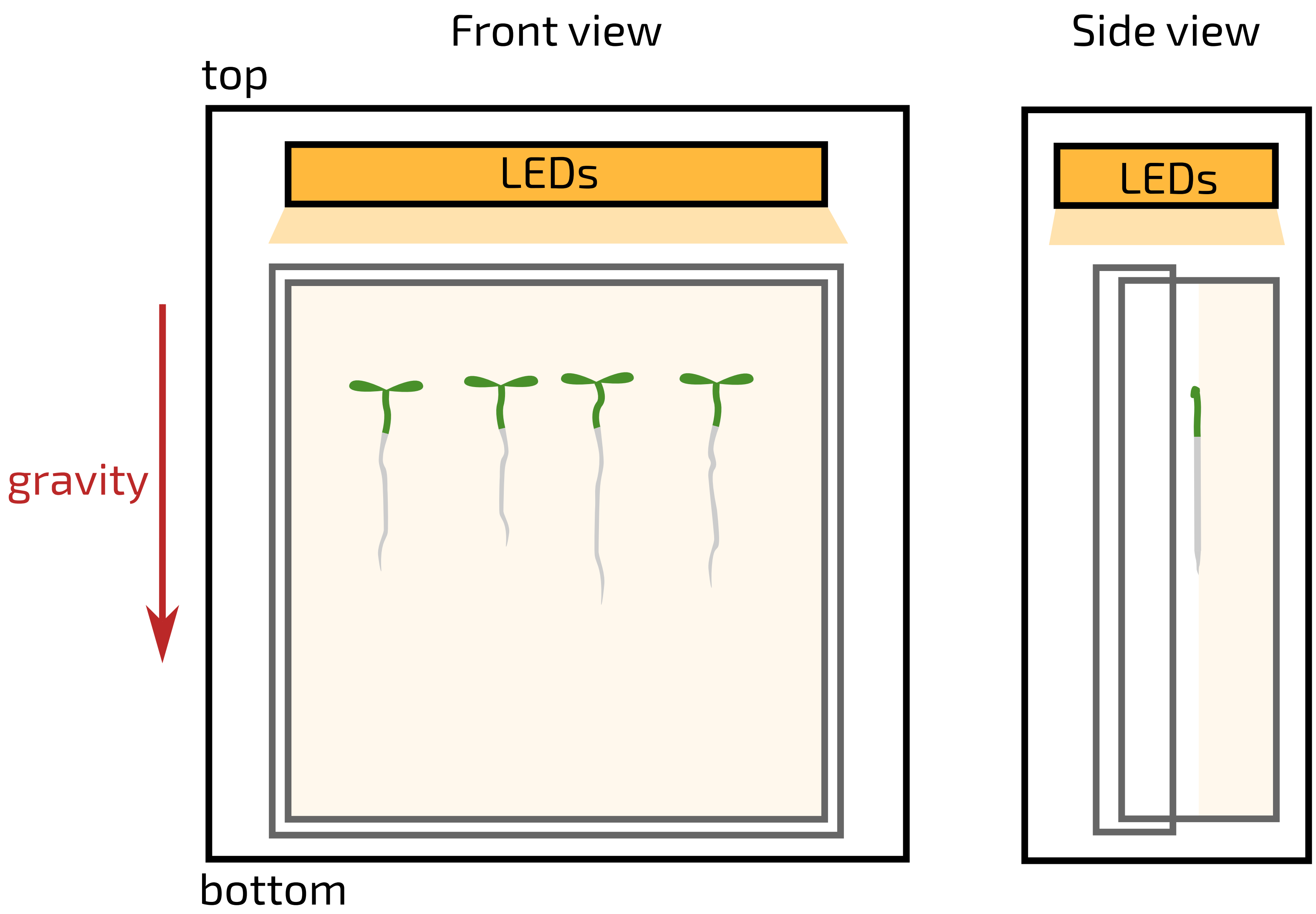


Figure 1: Diagram of the growth chamber setup. The "top" and "bottom" of the chamber and the orientation of the ground-based chamber relative to the gravity vector are indicated. Seedlings are shown growing as they would in the ground-based growth chamber. Note that the seedlings grew on the surface of the agar, as shown in the side view.

The following questions are based on some real experiments that were carried out, and some hypothetical ones.

Part A

Control variables

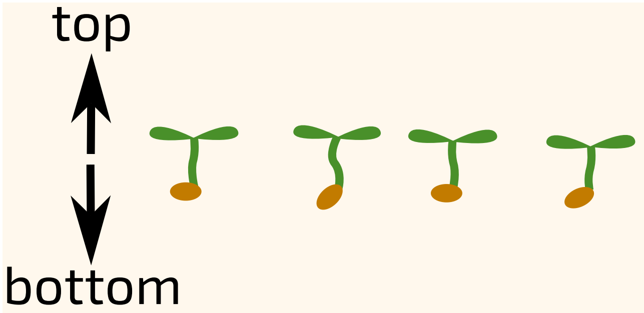
Which of the following variables should have been controlled during this investigation? Select all that apply.

- [ ] light intensity
- [ ] temperature
- [ ] humidity
- [ ] plant genotype
- [ ] gravitational environment
- [ ] plant phenotype
- [ ] growth medium

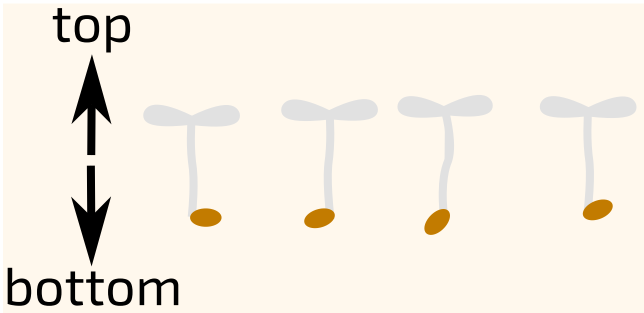
Part B

Shoot angle

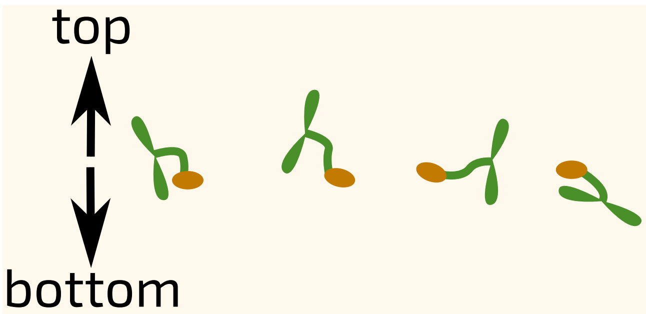
Which of the images below shows how the **shoots** grew on the ISS?



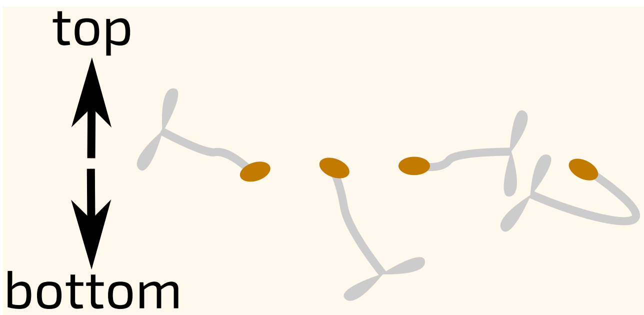
A



B



C



D

- A
- B
- C
- D

Part C

Root length

Figure 2 compares the length of roots and shoots in seedlings grown on the ground and on the ISS.

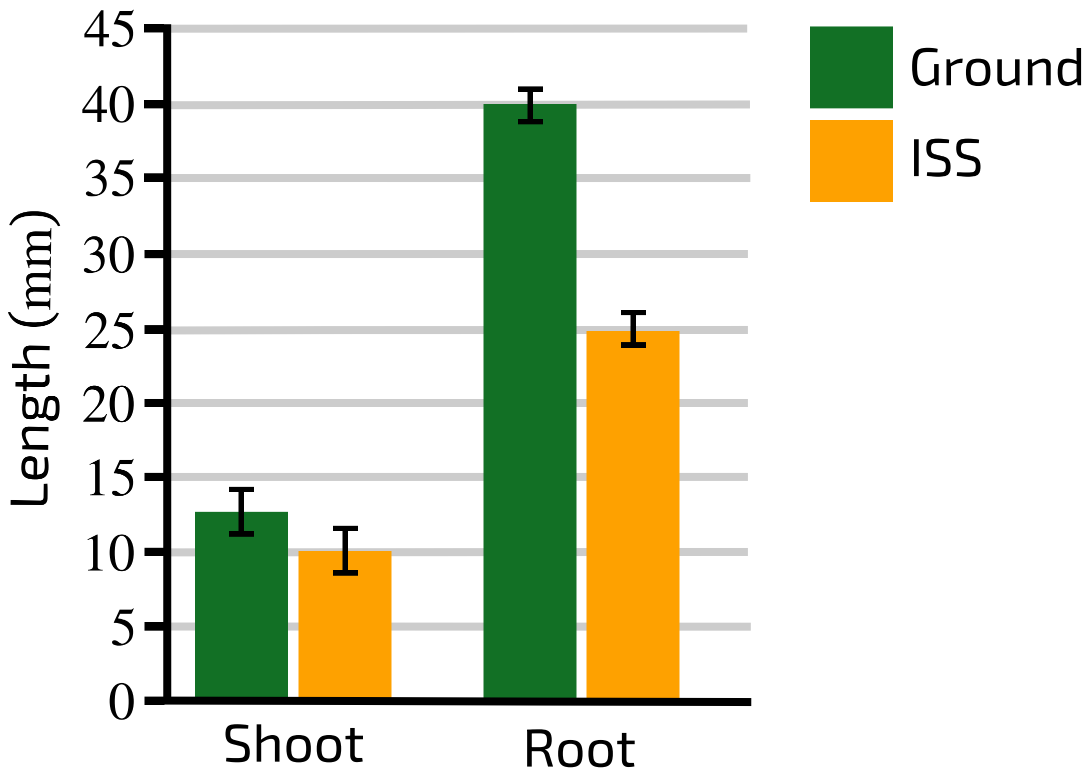


Figure 2: Root and shoot length of seedlings grown on the ground (green) and ISS (yellow) after 8 days of growth. Bar height shows the mean value; whiskers show the standard deviation. For each genotype n = 7.
Adapted from [1], CC BY 2.0

What was the percentage decrease in root length associated with growing plants on the ISS?

Which of the following are correct interpretations of this graph? Select all that apply.

- [ ] The mean **shoot** length is shorter on the ISS, and this difference is **likely** to be statistically significant.
- [ ] The mean **shoot** length is shorter on the ISS, and this difference is **unlikely** to be statistically significant.
- [ ] The mean **root** length is shorter on the ISS, and this difference is **likely** to be statistically significant.
- [ ] The mean **root** length is shorter on the ISS, and this difference is **unlikely** to be statistically significant.

Part D

Root angle

The researchers quantified the angle of root growth over time. At each time point, they measured the angle of the root tip relative to a vertical line. A measurement of 0° indicates the root tip was pointing straight down. A negative value indicates the root tip was pointing to the left, and positive value indicates the root tip was pointing to the right.

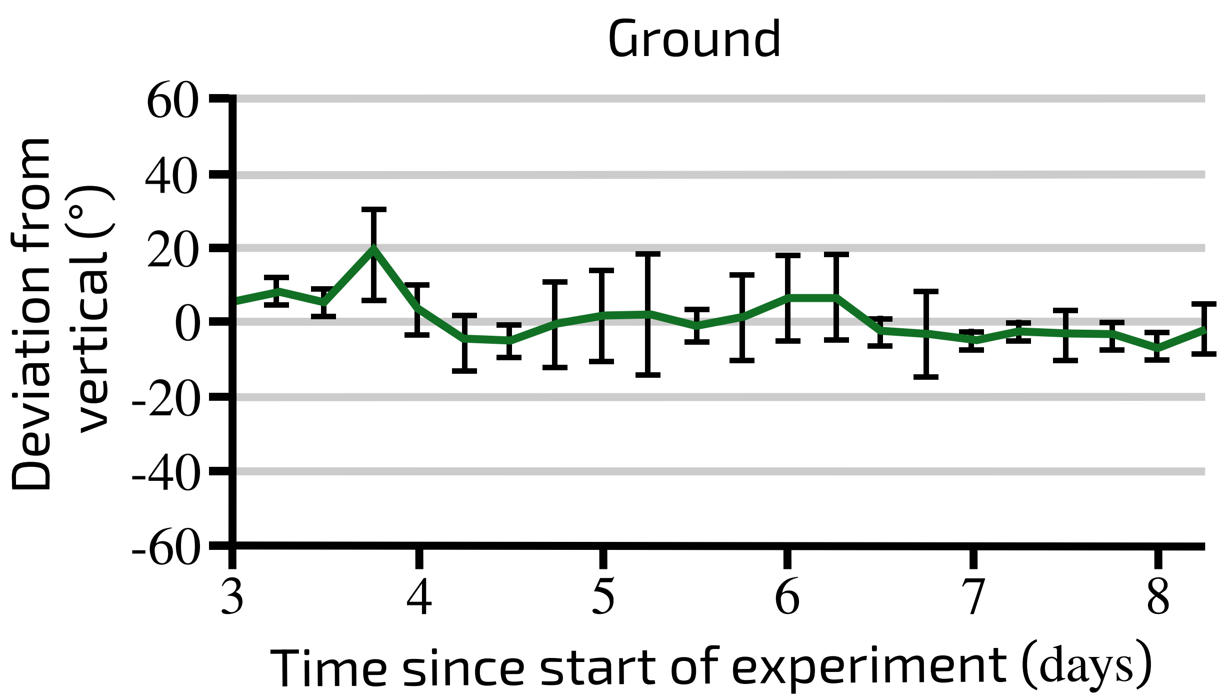


Figure 3: Green line shows average (mean of 3 samples) deviation from a vertical line. Error bars show standard deviation. Adapted from [1], CC BY 2.0

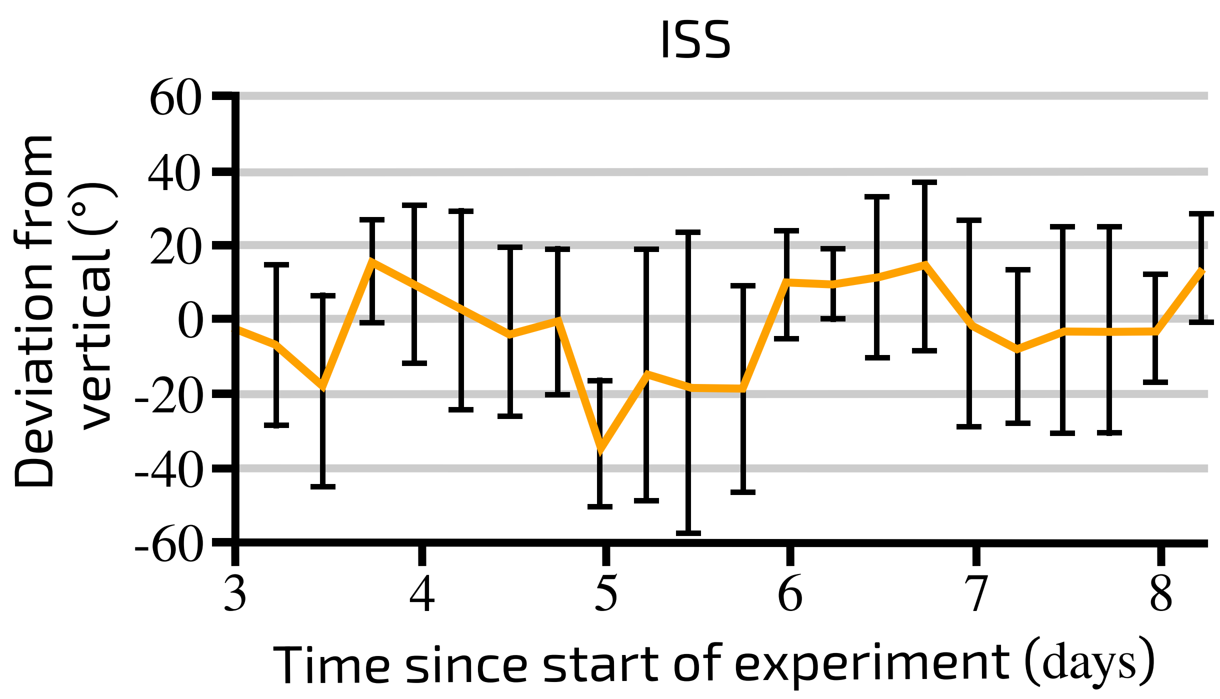


Figure 4: Yellow line shows average (mean of 4 samples) deviation from a vertical line. Error bars show standard deviation. Adapted from [1], CC BY 2.0

Which of the following statements are correct interpretations of these data? Select all that apply.

- [ ] At most time points (and overall), the root growth angle was more variable in **ISS**-grown plants.
- [ ] At most time points (and overall), the root growth angle was more variable in **ground**-grown plants.
- [ ] The root growth angle in ISS-grown plants was completely **random**.
- [ ] The overall growth direction of ISS-grown plants was **vertically down**.
- [ ] The overall growth direction of ISS-grown plants was strongly skewed to the **left**.

- [ ] The overall growth direction of ISS-grown plants was strongly skewed to the **right**.

Part E

Hormone distribution

The petri dishes were rotated 90°. At the same time, the LEDs were switched off (**Figure 5**).

Figure 6 shows the distribution of auxin in the root tips of plants grown on the ground and the ISS **before being rotated**. Note that the auxin distribution is the same in both.

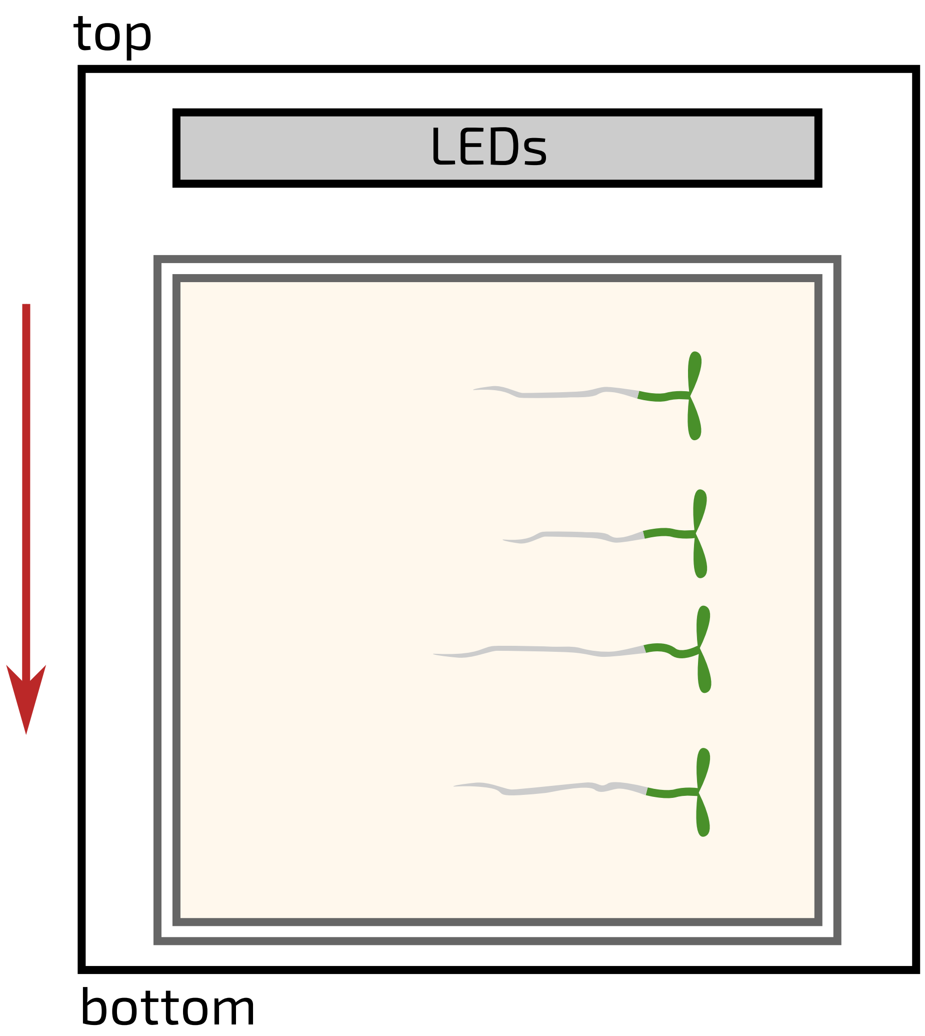


Figure 5: Diagram of the growth chamber setup. Seedlings are shown growing as they would in the ground-based growth chamber, immediately after rotation.

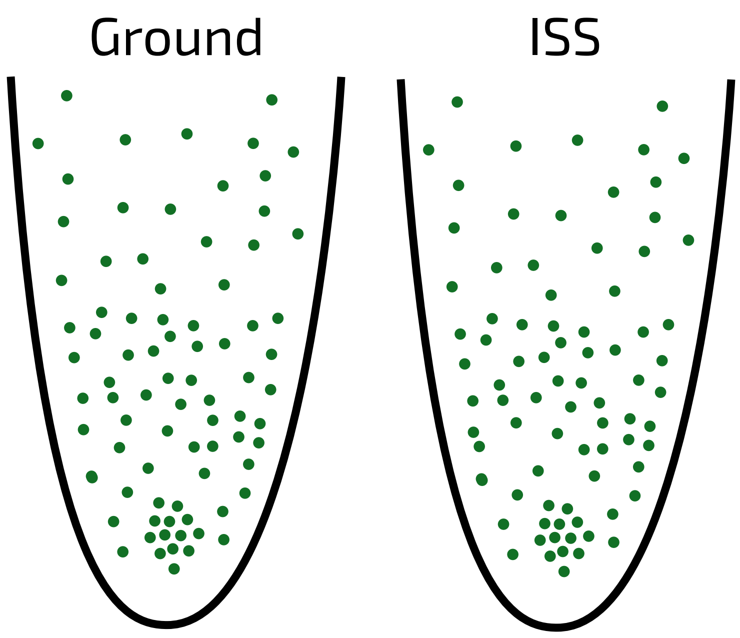
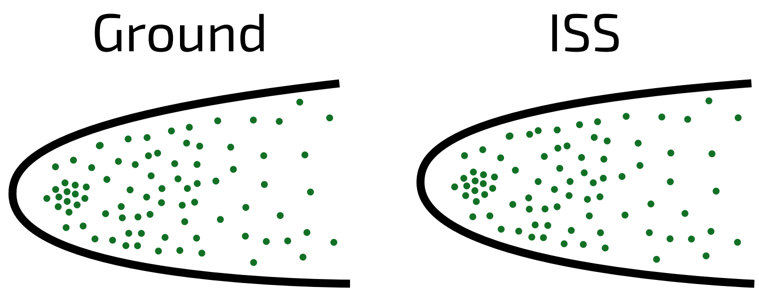
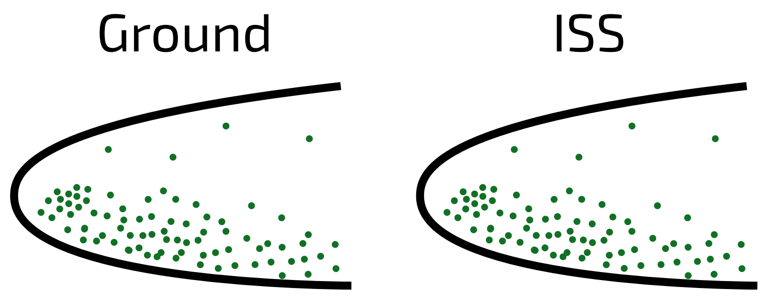


Figure 6: Auxin distribution in the root tips of ground- and ISS-grown seedlings.

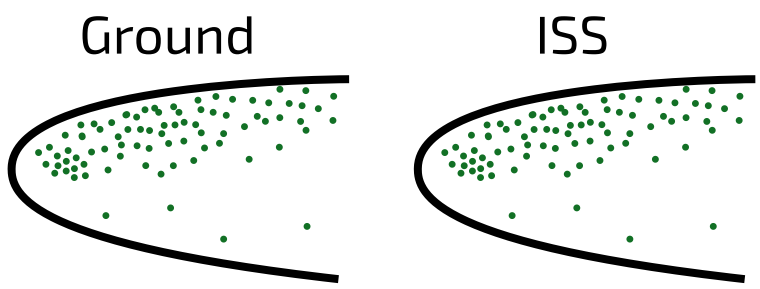
Which of the images below shows the distribution of auxin in the root tips several minutes **after** the seedlings were rotated?


Ground
ISS

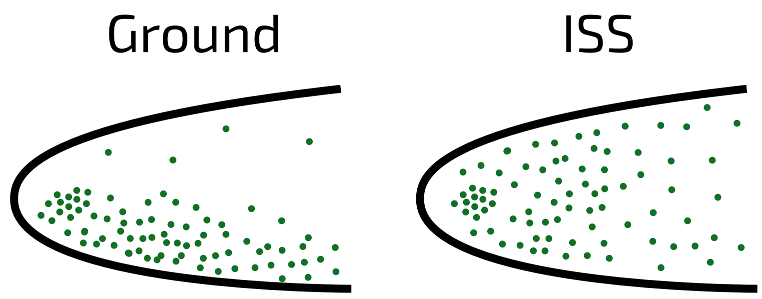
A


Ground
ISS

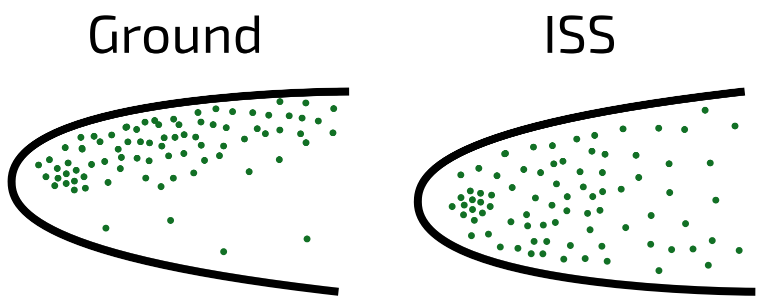
B


Ground
ISS

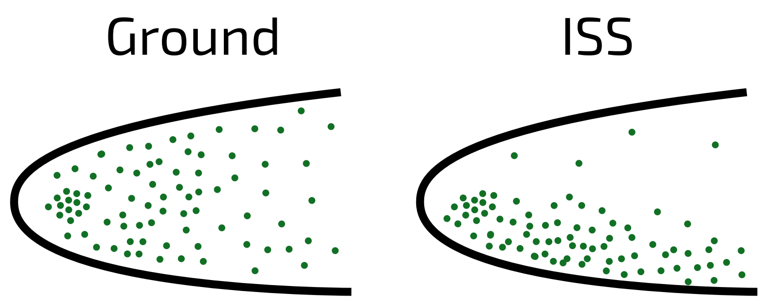
C


Ground
ISS

D


Ground
ISS

E


Ground
ISS

F

- A
- B
- C
- D
- E
- F

Fill in the blanks to describe how reorientation of the ground-based root causes a change in growth direction:

When the root is reoriented, gravity ______ to the ______ side of the root. The accumulation of auxin in this region ______ growth, while the low concentration on the ______ side of the root ______ growth. This causes the root to bend downwards.

Items:

pulls auxin | stimulates auxin transport | upper | lower | inhibits | promotes

1 Paul, AL., Amalfitano, C.E. & Ferl, R.J. Plant growth strategies are remodeled by spaceflight. BMC Plant Biol 12, 232 (2012). https://doi.org/10.1186/1471-2229-12-232

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Question deck:
STEM SMART Biology Week 21 - Plant Growth & Reproduction

Photoperiodism

Subject & topics: Biology | Physiology | Homeostasis & Response **Stage & difficulty:** A Level C2

Photoreceptors are light-sensitive proteins that regulate many aspects of plant development. Photoreceptors play important roles in the regulation of flowering, for example allowing plants to detect changes in day length and thereby coordinate flowering with changes in the seasons. This phenomenon is called photoperiodism.

Part A

Phytochromes

Phytochromes are a family of photoreceptors that are sensitive to red and far red light. Phytochromes exist in two forms P_R and P_{FR}. **Figure 1** shows how light affects the interconversion of phytochromes between these two forms.

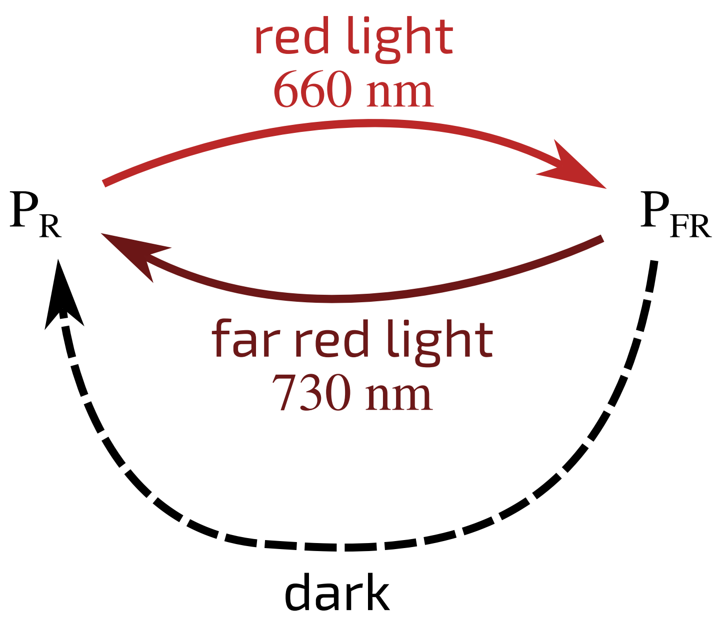


Figure 1: The interconversion of P_R and P_{FR}. Red light causes P_R to rapidly convert to P_{FR}. P_{FR} slowly converts back to P_R in the dark. Far red light causes P_{FR} to rapidly convert to P_R.

A plant species (species **X**) was grown under five different light regimes. **Figure 2** shows these light regimes and indicates whether the plants flowered under each regime. The white light used in this experiment contained a mixture of wavelengths that included red light but not far red light.

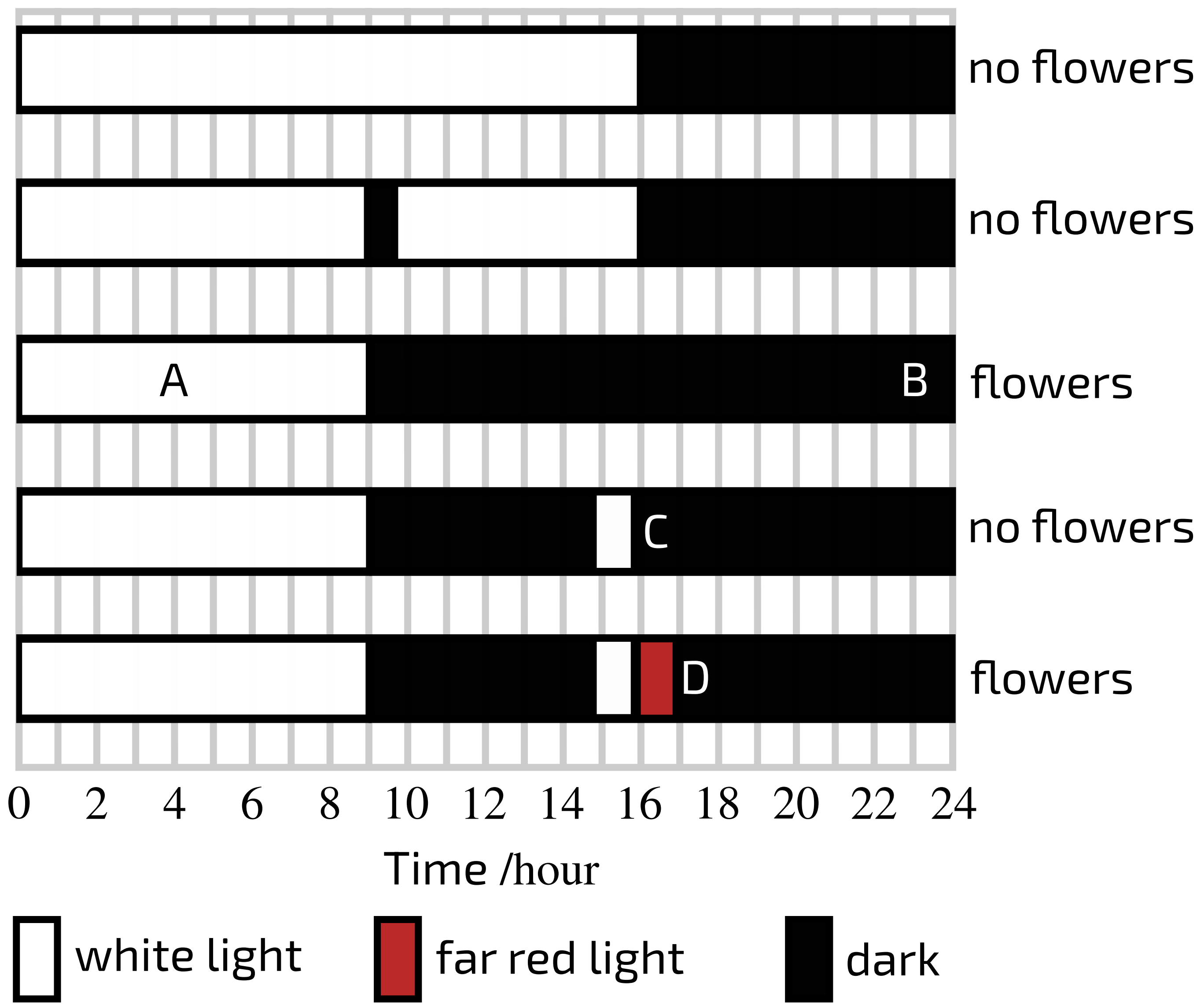


Figure 2: Flowering of species X under different light regimes. The letters A-D label time points that are referenced later.

Is species X a long day plant or a short day plant?

short day

long day

Fill in the table below to indicate the most abundant form of phytochrome present at time points A-D in **Figure 1**.

A	
B	
C	
D	

Items:

P_R P_{FR}

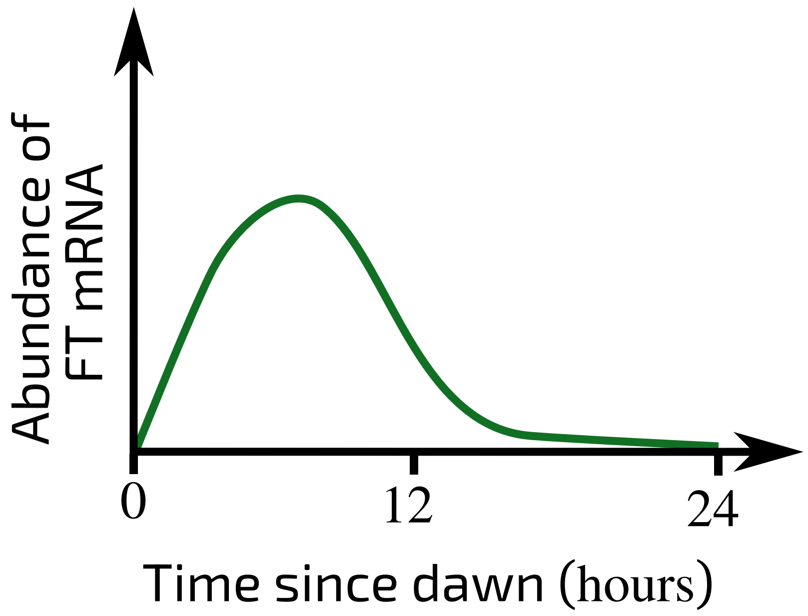
Part B

FT mRNA

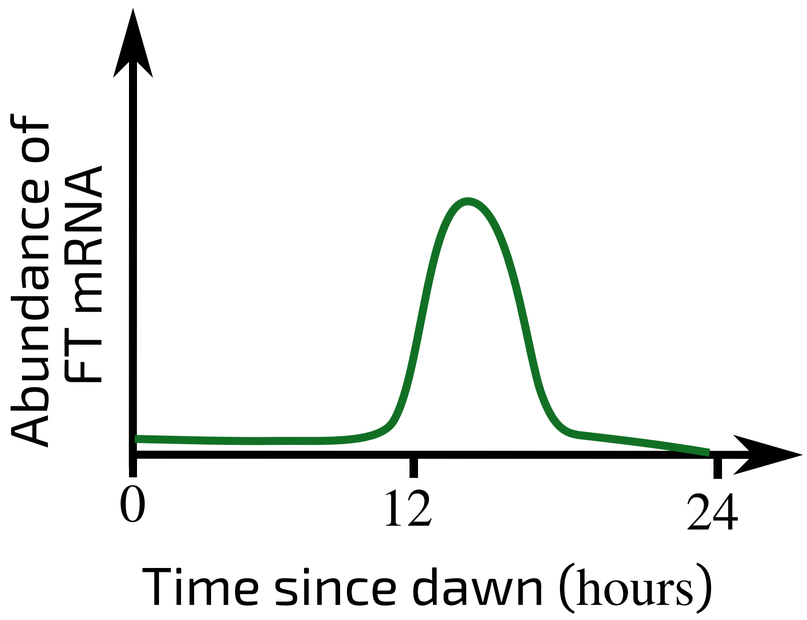
Flowering is stimulated by a protein called FT, which is primarily produced in leaves. In plant species **Y** (a long day plant), FT transcription is activated by a protein called CONSTANS.

CONSTANS transcription is regulated on a 24 hour cycle. During the ~12 hours after dawn, CONSTANS mRNA **is not** transcribed. During the ~12 hours preceding dawn, CONSTANS mRNA **is** transcribed. CONSTANS protein is stabilised by light-activated photoreceptors. In the dark (when the photoreceptors are inactive), CONSTANS protein is rapidly degraded, so that it cannot activate the transcription of FT.

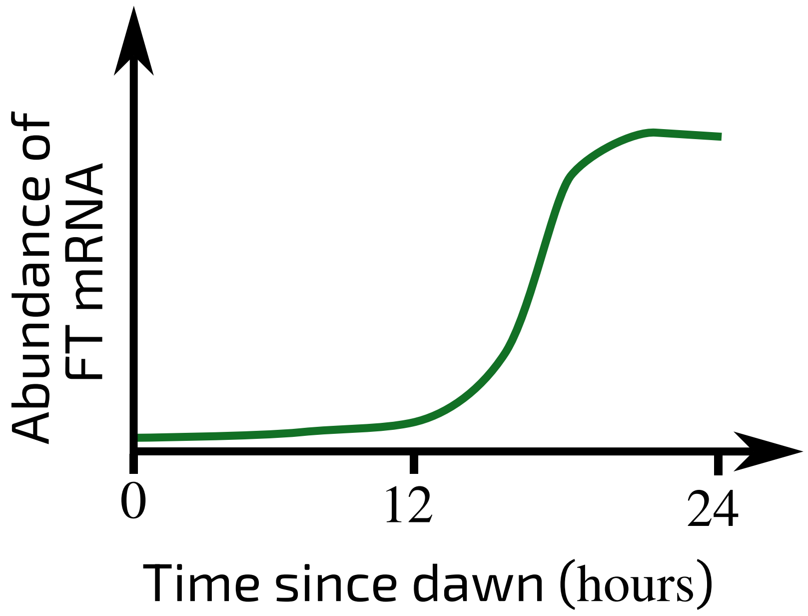
Plots A to F show hypothetical changes in FT mRNA levels over the course of 24 hrs.



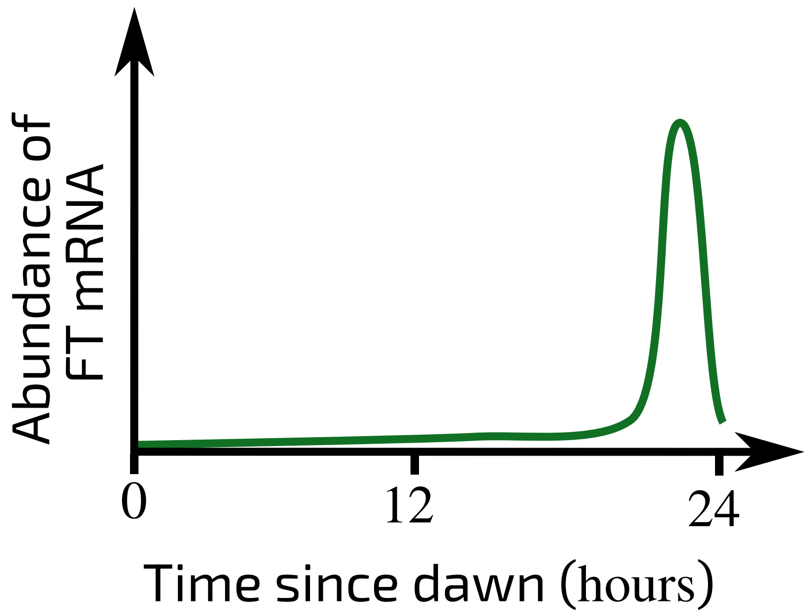
A



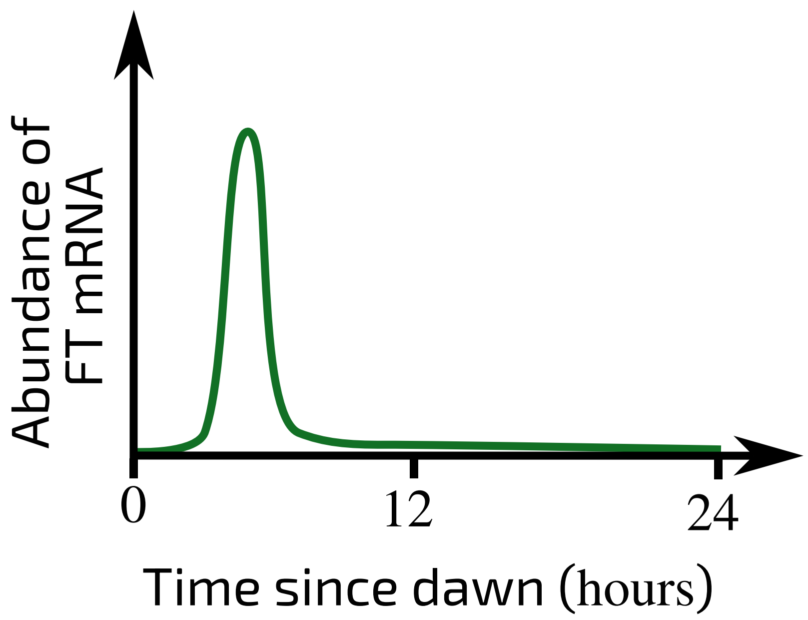
B


Abundance of FT mRNA
0
12
24
Time since dawn (hours)

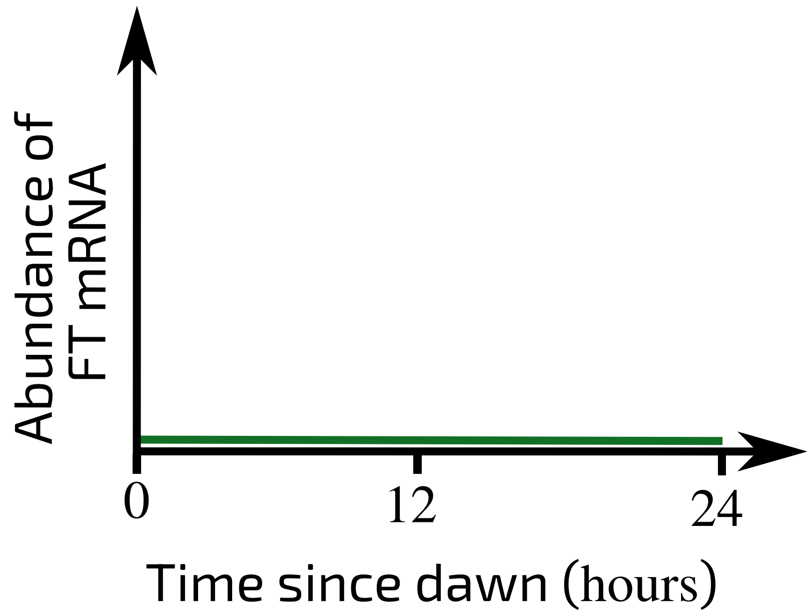
C


Abundance of FT mRNA
0
12
24
Time since dawn (hours)

D


Abundance of FT mRNA
0
12
24
Time since dawn (hours)

E



F

Which of the plots (A-F) show the level of FT mRNA in:

- species Y plants grown in **short day** conditions (9 hrs light: 15 hrs dark)? ☐
- species Y plants grown in **long day** conditions (16 hrs light: 8 hrs dark)? ☐

Items:

A B C D E F

Part C

Grafting experiment

A species X stem was grafted onto a species Y rootstock (roots, lower stem and branches). The grafted plant was grown in long day conditions ($16\,\mathrm{hrs}$ light: $8\,\mathrm{hrs}$ dark).

Which of the following statements describe the most likely outcome(s) of this graft? Select all that apply.

- [ ] FT mRNA or protein will move from the species Y portion to the species X portion.
- [ ] FT mRNA or protein will move from the species X portion to the species Y portion.
- [ ] FT will not be expressed anywhere in the grafted plant.
- [ ] Only the species X branches will flower.
- [ ] Only the species Y branches will flower.
- [ ] Both the species X and species Y branches will flower.
- [ ] None of the branches will flower.

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Question deck:

STEM SMART Biology Week 21 - Plant Growth & Reproduction

STEM SMART Biology Week 21 - Plant Growth & Reproduction >

Angiosperm Reproduction

Subject & topics: Biology | Physiology | Reproduction & Development **Stage & difficulty:** A Level P2

The angiosperms are the group of plants that produce flowers. In most species, each flower contains both male and female gametes. The figure below shows the structure of a flower in cross-section and from above.

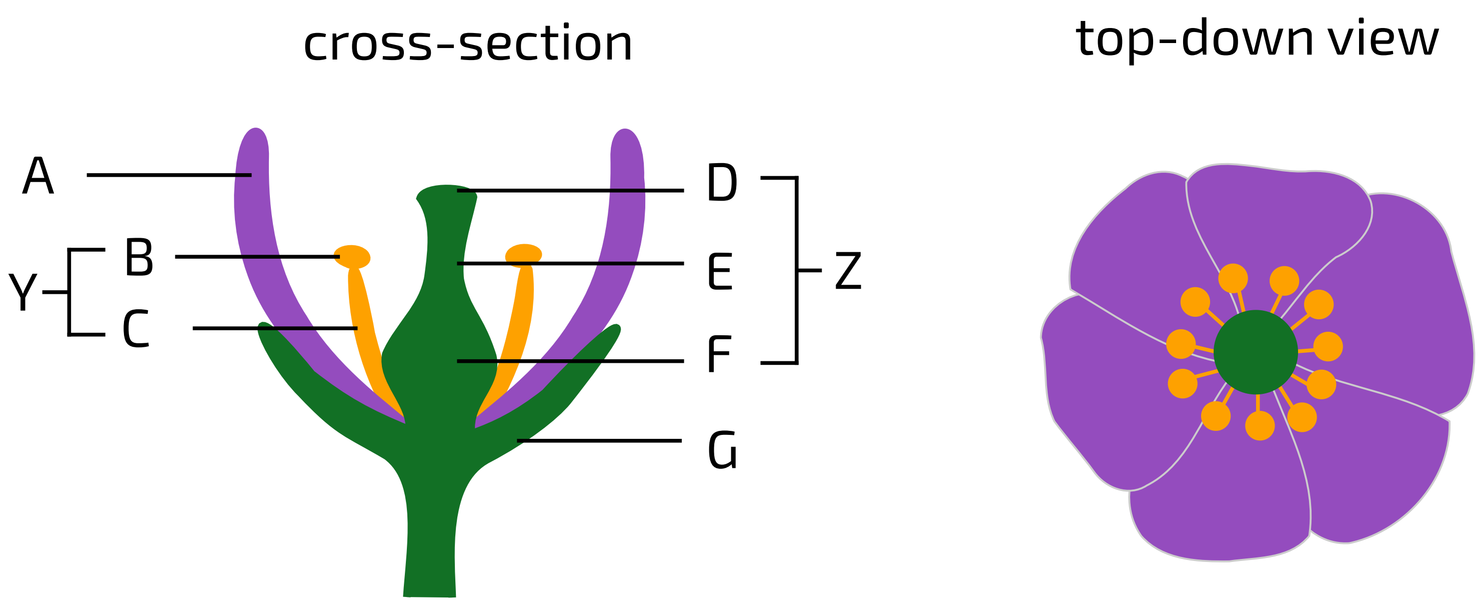

Figure 1: Structure of a flower. Left: cross-section side view, with different structures labelled. Right: the same flower from a top-down view.

Part A

Flower anatomy

Match the structures to the figure labels in the table below.

Label	Structure
A	
B	
C	
D	
E	
F	
G	
Y	
Z	

Items:

anther carpel filament ovary petal sepal stamen stigma style

Part B
Flower functions

Match the structures to the functions in the table below.

Function	Label
site of pollen production	
site of pollen deposition	
site of ovules	
brightly-coloured to attract pollinators	
protects the flower before it opens	

Items:

A B C D E F G

Part C
Pollination

Pollination is the process by which pollen (which contains male gametes) is transferred to the stigma - usually of a flower of a different plant. In most angiosperms, this is done by insect pollinators. These insects receive nectar from the flowers.

What is the name for this type of interaction between two organisms, which is beneficial for both?

Part D

Double fertilisation

When a pollen grain lands on the stigma of another flower, it forms a tube that grows down through the style and into the ovary. The pollen grain contains two haploid sperm cells. Both of these travel down the pollen tube into an ovule. One sperm cell fertilises the haploid egg cell to produce the diploid zygote. The other sperm cell fertilises **two** haploid "polar nuclei" in the ovule to produce a triploid endosperm. The endosperm protects the developing plant embryo and provides it with nutrition.

This whole process, which is unique to angiosperms, is called double fertilisation.

Peas (*Pisum sativum*) are angiosperms with a diploid chromosome number of .

How many chromosomes does a pea sperm cell have?

How many chromosomes does a pea zygote have?

How many chromosomes does a pea endosperm cell have?

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Question deck:

STEM SMART Biology Week 21 - Plant Growth & Reproduction

STEM SMART Biology Week 21 - Plant Growth & Reproduction >

The ABC Model

Subject & topics: Biology | Physiology | Reproduction & Development **Stage & difficulty:** A Level C2

The size, colour, shape and number of floral organs varies considerably between the flowers of different species. However, the overall **organisation** of a flower is very **consistent** between species.

A typical flower is organised into four concentric whorls (i.e. rings/circles) of organs (**Figure 1**). Whorl 1 (the outermost whorl) is the sepals, the whorl 2 is the petals, whorl 3 is the stamens and whorl 4 (in the middle of the flower) is the carpel.

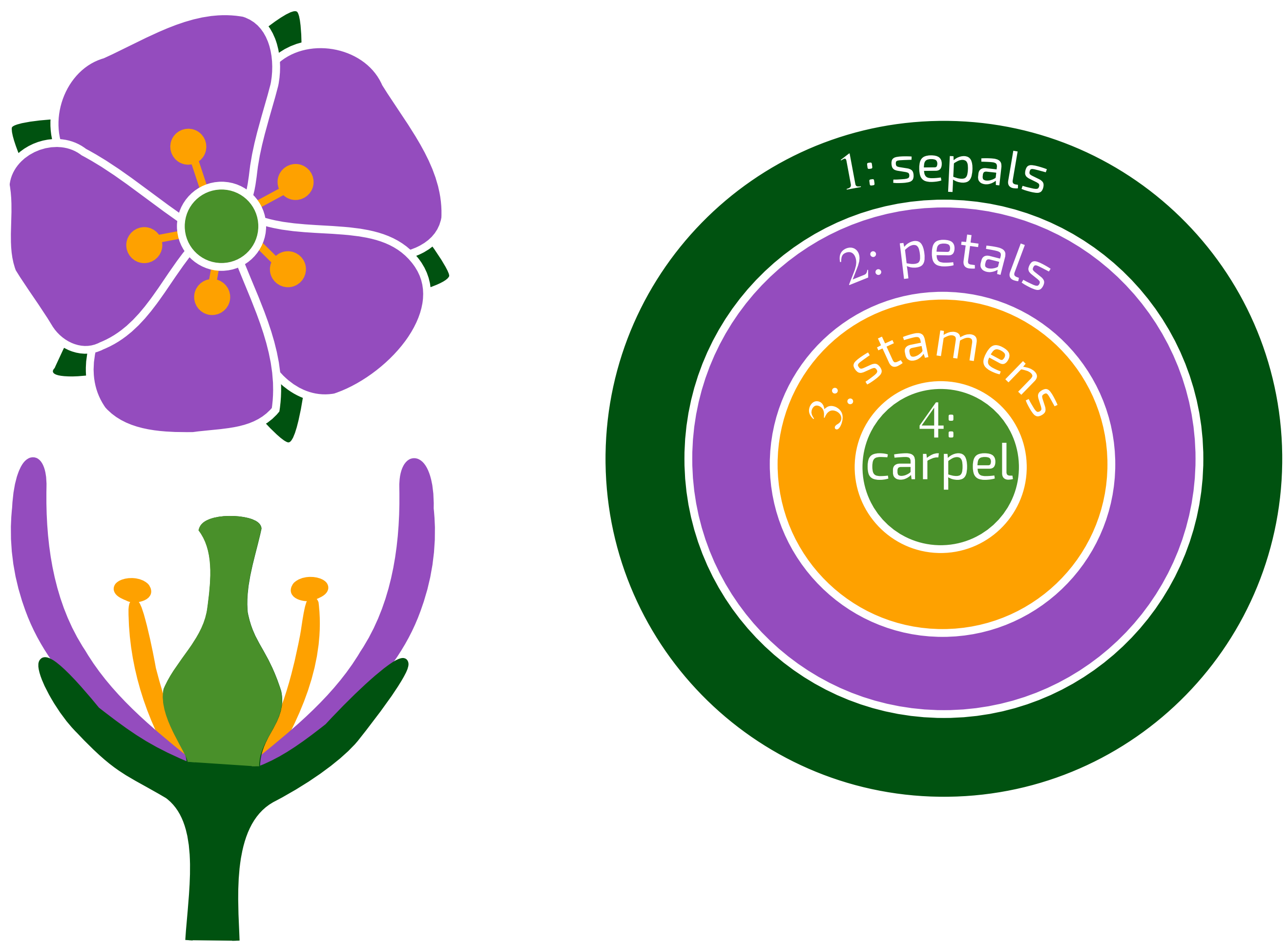

Figure 1: Organisation of a flower. Top left: top-down view of a flower. Bottom left: cross-section through a flower. Right: the organisation of floral organs into four concentric whorls (rings).

A set of genes — called A, B, and C genes — are responsible for this organisation. These genes are expressed in different regions of the flower (**Figure 2**). A protein-coding gene is said to be expressed when its mRNA is actively being transcribed and translated.

- The A gene is expressed in whorls 1 and 2.
- The B genes are expressed in whorls 2 and 3.
- The C gene is expressed in whorls 3 and 4.

The expression of these genes in a particular region of the flower determines which floral organs develop:

- Expression of the A gene by itself produces sepals.
- Expression of A and B genes together produces petals.
- Expression of B and C genes together produces stamens.
- Expression of the C gene by itself produces carpels.

Changing the expression of the A, B or C genes alters the development of the flower.

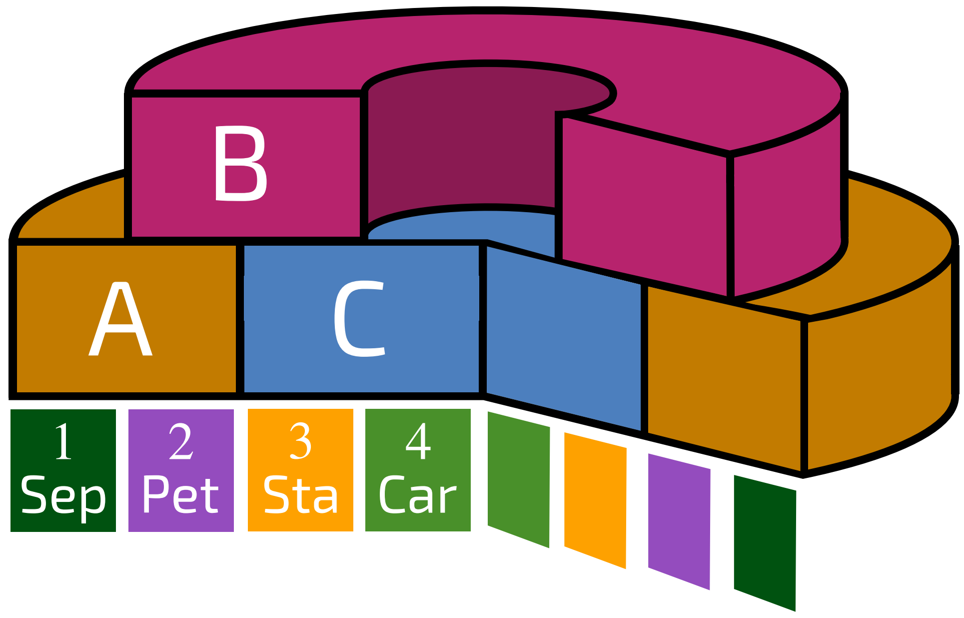

Figure 2: The overlapping expression of A, B, and C genes in the flower.

Part A

B mutants

The B genes in a plant were mutated so that they no longer functioned. Predict the arrangement of floral organs in this mutant.

Whorl 1	Whorl 2	Whorl 3	Whorl 4

Items:

sepals petals stamen carpel

Part B

A mutants

The C gene inhibits the expression of the A gene, and the A gene inhibits the expression of the C gene. This is why the A and C genes are expressed in different regions of the flower. However, if either gene is deactivated, the other gene becomes expressed throughout the whole flower.

The A gene in a plant was mutated so that it no longer functioned. Predict the arrangement of floral organs in this mutant.

Whorl 1	Whorl 2	Whorl 3	Whorl 4

Items:

sepals petals stamen carpel

Part C

C mutants

The C gene in a plant was mutated so that it no longer functioned. Predict the arrangement of floral organs in this mutant.

Whorl 1	Whorl 2	Whorl 3	Whorl 4

Items:

sepals petals stamen carpel

Part D

Double mutants

Predict the arrangement of floral organs in plants that have the following combinations of mutations.

Mutations	Whorl 1	Whorl 2	Whorl 3	Whorl 4
B and A				
B and C				

Items:

sepals petals stamens carpel

Part E

E genes

A fourth class of genes, called the E genes, are also required for floral organ development. E genes are expressed in whorls 2, 3, and 4. E genes are required for B and C genes to function, including the ability of C genes to inhibit A genes.

The E genes in a plant were mutated so that they no longer functioned. Predict the arrangement of floral organs in this mutant.

Whorl 1	Whorl 2	Whorl 3	Whorl 4

Items:

sepals petals stamen carpel

Part F

Leaf expression

A, B, C and E genes are not expressed in leaves.

A scientist genetically modified a plant so that it simultaneously expressed B and C genes in its leaves. How did this affect leaf development?

- The leaves developed into sepals.
- The leaves developed into petals.
- The leaves developed into stamens.
- The leaves developed into carpels.
- Leaf development was unchanged.

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